Artificial Intelligence Technology and Application

Python
Lab Guide

Student Version

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1 Treasury Management System

1.1 Introduction

1.1.1 About This Lab

Python is used to implement a treasury management system, which provides deposit, withdrawal, transfer, secret management, and credential generation functions. Data is stored in the MySQL database.

1.1.2 Objectives

For the comprehensive application of Python basic syntax and advanced syntax, a simple treasury management system is implemented.

1.2 Procedure

1.2.1 Experiment Approach

Use the PyMySql to connect to and operate the database, and log in to the database to determine the information in the database. After the login is successful, the system welcome page is displayed. In addition, a user object is created for the user who logs in successfully. The corresponding method is performed according to the operation performed by the user, and the operation is synchronized to the database. After the operation is complete, the operation is recorded, that is, the operation is written to a local file.

1.2.2 Implementation

Step 1 Create a database and data table.

Create a database.

Create a data table.

Insert data.

```
```

Considering that the system connects to the database for many times and the statements for connecting to the database are similar, the database connection is encapsulated as a method.

```
# Define the method of connecting to the database. The SQL statement is the database operation statement to be executed each time.
def con_mysql(sql):
```

Test method:

```
sql = "select * from user"
con_mysql(sql)
```

Output:

```
('admin',
 '123456',
 datetime.datetime(2019, 4, 23, 12, 15, 21),
 datetime.datetime(2019, 4, 23, 12, 15, 25),
 11200.0)
```

Figure 1-1 Database connection test result

Step 2 Define a user class.

```
class Account(object):
```

```
# Account balance.

# User name.

      # Last login time.


     # Deposit.
     def save(self):


     # Withdrawal.
     def take(self):



     # Change the password.
     def update(self):


     # Transfer.
     def transfer(self):



     # Perform the selected operation.




# Generate the credential after the operation.
     def voucher(self,end_time, action):


Login function
```

Test the login function:

```
user_account = login()
user_account
```

Output:

```
Please enter your account:admin
Please enter your password:
```

Figure 1-2 Login function test

Step 3 Welcome window

```
def welcome():
```

Test the welcome method:

```
action = welcome()
action
```

Output:

```
* * * * * * * * * * * * * * * * * * * * * * * * *
*                                               *
*     Welcome to the Fund Management System     *
*                                               *
* * * * * * * * * * * * * * * * * * * * * * * * *
Please select the operation: 1. Deposit 2. Withdraw 3. Transfer 4. Change Password 5. Exit:
```

Figure 1-3 Welcome window

Step 4 Define the system startup function.

Configure the startup function.

```
def run():
```

Define the decorator.

```
def consume_time(func, *args, **kwargs):
    def inner(*args, **kwargs):
```

Add the following function to the system startup function:

```
@consume_time
def run():
```

Output:

```
Please enter your account:admin
Please enter your password:********
Current login time2021-04-29 09:50:48
 * * * * * * * * * * * * * * * * * * * * * * * * *
 *                                               *
 *     Welcome to the Fund Management System     *
 *                                               *
 * * * * * * * * * * * * * * * * * * * * * * * * *
Please select the operation: 1. Deposit 2. Withdraw 3. Transfer 4. Change Passw
ord 5. Exit:1
Please enter the deposit amount:100
Saved CNY 100.000000
Generation succeeded. Keep it safely
Logout time 2021-04-29 09:51:24
```

Figure 1-4 System execution result

```
account:admin
operation: Deposit
deposit amount:100
Current login time2021-04-29 09:50:48
Logout time 2021-04-29 09:51:24
```

Figure 1-5 Generated credential